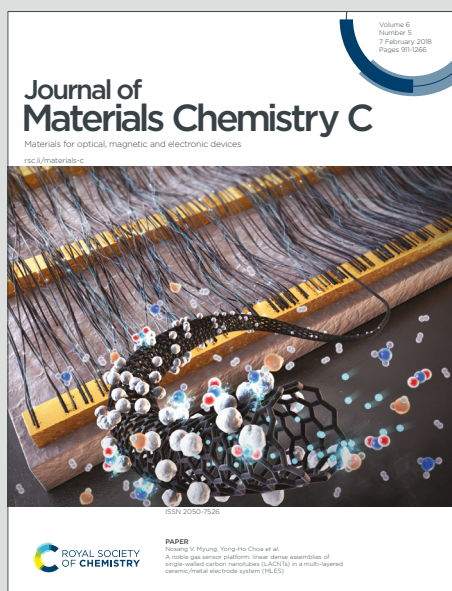


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Critical Evaluation of the Thermometric Performance of Ratiometric Luminescence Thermometers based on $\text{Ba}_3(\text{VO}_4)_2\text{:Mn}^{5+}$, Nd^{3+} for Deep-Tissue Thermal Imaging

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Abstract



Near-infrared (NIR) luminescence thermometry has been brought to the fore as a reliable approach for remote thermal sensing and imaging. Lanthanide (Ln^{3+})-based nanophosphors are often proposed as NIR nanothermometers of choice. However, the combination of Ln^{3+} with transition metal (TM) ions has recently emerged as a strategy to introduce additional emission bands and/or $\text{TM} \leftrightarrow \text{Ln}^{3+}$ energy transfer pathways whose temperature dependence can be harnessed to increase the sensitivity of the thermometric approach. Yet, the examples of combination of luminescent nanothermometers working in the NIR and hosting simultaneously TM and Ln^{3+} are scarce, leaving plenty of space to the exploration of these systems. Herein, we report on the preparation and optimization of the thermometric performance of $\text{Ba}_3(\text{VO}_4)_2\text{Mn}^{5+}, \text{Nd}^{3+}$ nanophosphors. The different temperature dependence of the emission intensity of the two doped luminescent centers allows using the ratio between Mn^{5+} and Nd^{3+} as a reliable thermometric parameter with relative thermal sensitivities of $1\% \text{K}^{-1}$ close to room temperature. We then showcase the suitability of this nanophosphor to be employed for 2D NIR luminescence thermal imaging. Lastly, we critically evaluate the possibility to use this thermal imaging approach through opaque media with the help of phantoms with tissue-like optical properties. As expected, a loss of reliability of the thermometric method is observed due to tissue-induced photon scattering and absorption that differentially affect the emission of Mn^{5+} and Nd^{3+} . Overall, the reported results underscore the good performance of the newly developed nanothermometer, while consolidating the call for the use of luminescent nanothermometers working in the time-domain (rather than in the spectral domain) for deep-tissue thermal readout/imaging.

Introduction

Thermal sensing and thermal imaging with near-infrared (NIR) emitting luminescent thermometers are particularly appealing from a biological standpoint^{1–4}. The ability to monitor



the temperature of biological systems remotely and in a minimal invasive way has opened up new research avenues, allowing to reach a better understanding of biological processes at the tissue and cellular levels, as well as the development of highly specific diagnostic tools^{5,6}. Deep-tissue thermal imaging requires the use of phosphors with high luminescence brightness in the optical biological windows (750-950 nm (NIR-I), 1000-1350 nm (NIR-II), and 1500-1800 nm (NIR-III), wherein the scattering and absorption of light by tissues is reduced^{7,8}.

One of the most reliable methods of measuring the temperature in luminescence thermometry is a ratiometric approach employing the ratio of the intensities of two emission bands with different temperature-induced variations⁹⁻¹⁴. A significant difference in the thermal dependence of the two bands under consideration ensures a high measurement sensitivity in this approach. Co-doping transition metal (TM) ions and lanthanide (Ln^{3+}) ions in the same matrix, for example, provides such a possibility¹⁴⁻²¹. Temperature-induced luminescence quenching mechanisms differ depending on the configuration of the energy levels of both types of dopants²². With proper ion selection, the Ln^{3+} signal (with less thermal variation) can be used as an internal luminescence reference.

In this vein, recent studies have pointed at Mn^{5+} as an ideal TM for the preparation of luminescent thermometers based on TM and Ln^{3+} ²³⁻²⁷. Indeed, its emission band in the NIR-II is highly susceptible to thermal quenching. Its spectrally narrow emission also limits overlaps with the NIR emission of co-doped Ln^{3+} . This aspect allays concerns over loss of reliability of ratiometric luminescence thermometry arising from signal admixing. However, the literature on the thermometric performance of Mn^{5+} -based luminescent thermometers is scarce, and even fewer studies deal with thermal imaging using this class of phosphors²³⁻²⁷. Given the above, the goal of this work is to critically evaluate the sensing and thermal imaging potential of $\text{Ba}_3(\text{VO}_4)_2:\text{Mn}^{5+}$, Nd^{3+} based on the ratio of Mn^{5+} and Nd^{3+} ion intensities.



In this paper, we present an unprecedented $\text{Ba}_3(\text{VO}_4)_2\text{Mn}^{5+}, \text{Nd}^{3+}$ luminescent thermometer. The luminescence intensity ratio (LIR) between Mn^{5+} and Nd^{3+} emissions provides a reliable thermometric parameter. Subsequently, the concentration of Nd^{3+} ions was optimized to maximize the thermal sensing capability of the developed material. Following that, 2D temperature imaging of a thermal gradient was performed through glass first and then through tissue phantoms, simulating the difficult situation encountered in *in vivo* thermal sensing. A critical evaluation of the performance of this ratiometric luminescence thermometry approach is provided, highlighting the difficulties that spectral-based methods inherently present.

Experimental

Materials and chemicals

The $\text{Ba}_3(\text{VO}_4)_2:1\% \text{Mn}^{5+}, y\% \text{Nd}^{3+}$ ($y = 0.1; 0.2; 0.5; 1; 2; 5$) nanocrystals were synthesised with a modified Pechini method²⁸. The following materials were used as reagents without further purification: $\text{Ba}(\text{NO}_3)_2$ (99.999% purity, Alfa Aesar), NH_4VO_3 (99.999% purity, Alfa Aesar), $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$ (>99.0% purity, Sigma Aldrich), Nd_2O_3 (99.99% purity, Stanford Materials Corporation) and $\text{C}_6\text{H}_8\text{O}_7$ (>99.5% purity, Alfa Aesar). The starting materials were weighed so that the total amount of moles of metals was equal to 4.00×10^{-4} mol.

Nanocrystals preparation

Stoichiometric amounts of $\text{Ba}(\text{NO}_3)_2$, NH_4VO_3 and $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$ were dissolved in deionized water in separate glasses and then mixed together. Neodymium oxide was dissolved in deionized water with the addition of 2 mL of HNO_3 (65% purity, Avantor), then recrystallized three times to remove the excess HNO_3 and added to the aqueous solution of other reagents. After that, 24.0×10^{-4} mol of anhydrous citric acid was added to the mixture with the molar ratio of citric acid at all metals set to 6:1. Subsequently, the samples with a concentration of 0.1%



Mn⁵⁺ in respect to the number of V⁵⁺ ions and y% Nd³⁺ in relation to the number of Ba²⁺ ions

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were calcined in porcelain crucibles at 1000 °C in air for 8 h. Finally, the obtained powders were ground in an agate mortar.

Tissue phantom preparation

Tissue phantoms were prepared based on a modified procedure given in Ref ²⁹. In this work, two types of phantom tissues were used, consisting of 50% Intralipids emulsion (tissue phantom A), and 50% Intralipids emulsion with 0.1% of Indian ink (tissue phantom B). For this purpose, the Intralipids emulsion (20% in water, Sigma Aldrich) and deionized water in a 1:1 ratio were introduced into a 100 mL Erlenmeyer flask to reach a final volume of 50 mL. The mixture was placed into an oil bath preheated at 363 K and kept under stirring (350 rpm). When considering tissue phantom B, 0.1% Indian ink was added at this stage. Next, 2%wt of agar powder was added and left under stirring at 363 K for 30 minutes until the agar was completely dissolved. After this time, the mixture was poured into a crystallizing dish and let cool down to room temperature until the gel set. All gels of 1.3 mm in thickness were stored in the fridge for further use.

Characterization

All synthesized materials were examined by X-ray powder diffraction (XRPD) measurements carried out on a PANalytical X'Pert diffractometer, equipped with an Anton Paar TCU 1000 N temperature control unit, using Ni-filtered Cu-K_α radiation (V = 40 kV, I = 30 mA).

Transmission electron microscope (TEM) images were taken using a FEI TECNAI G2 X-TWIN microscope. The powders were dispersed in methanol with the aid of ultrasounds and deposited on lacey-type copper grids. The studies were performed with 300 keV parallel beam electron energy. The images were digitally recorded using a Gatan Ultrascan 1000XP.



The emission spectra were measured using the 808 nm excitation lines from a laser diode (Changchun New Industries Optoelectronics Technology Co. Ltd.) and a NIRQUEST spectrometer from Ocean Optics (1.5 nm spectral resolution) as a detector. The excitation spectra and luminescence decay profiles were recorded using the FLS1000 Fluorescence spectrometer from Edinburgh Instruments with a R5509-72 photomultiplier tube from Hamamatsu in nitrogen-flow cooled housing as a detector with a 450 W halogen lamp and 668 nm pulsed work laser diode (Changchun New Industries Optoelectronics Technology Co. Ltd.) as excitation sources. The low-temperature emission spectra were also obtained with the FLS1000 fluorescence spectrometer. The temperature of the sample was controlled using a THMS 600 heating-cooling stage from Linkam (0.1 K temperature stability and 0.1 K set point resolution).

The quantitative composition of $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}, y\% \text{Nd}^{3+}$ ($y = 0.1; 0.2; 0.5; 1; 2; 5$) nanocrystals was determined using inductively coupled plasma atomic emission spectroscopy (ICP-AES). ICP-OES analysis was carried out using iCAP 7400 DUO ICP-OES Thermo Scientific spectrometer equipped with a high-performance solid-state Charge Injection Device (CID) detector, the CID86, enabling free choice of wavelengths over the 166 – 847 nm range. The experimental data were collected using Qtegra ISDS software. For measurement all the powders of $\text{Ba}(\text{VO}_4)_2: \text{Mn}^{5+}, \text{Nd}^{3+}$ were dissolved in 65% HNO_3 .

For imaging experiments, the sample was placed in an optical setup especially designed for fluorescence imaging in NIR-II: the PhotonSWIR Imager from BioSpace Lab (Nesles-la-Vallée, France). An InGaAs CCD camera (WiDy SenS 640V-ST, New Imaging Technologies, France) with a fiber-coupled 808 nm laser diode (LIMO) and two filters placed in an automatic filter wheel, namely a 1200 nm bandpass filter (Edmund Optics, Techspec OD4.0 25 nm Bandpass Filter FWHM = 25 nm) and a 1150 nm shortpass filter (Edmund Optics, Techspec OD2.0 Short-pass Filter) are the main components. Additionally, the camera is always protected



by an 850 nm long-pass filter (Thorlabs, FEL850). For the first experiment without the presence of tissue phantom and the second with tissue phantom A, the laser was set to a maximum output power of 14.1 mW and collected by a lens for illumination of the samples with a maximum (peak) power density of 0.05 mW/cm². For measurements via tissue phantom B, the maximum output power of the laser was set to 53.9 mW, resulting in a maximum power density of 0.21 mW/cm². The materials were illuminated with a laser diode in continuous mode, and the system was operated in ratiometric mode taking 25 image pairs with 0.5 s exposure time for each image and the automatic filter change taking 0.1 s in between, followed by a pause of 1 s before repeating the sequence with the next pair.

Results and discussion

Structural and morphological characterization

The rhombohedral structure of Ba₃(VO₄)₂ (with *R*-3*m* space group) consists of (VO₄)³⁻ tetrahedra and Ba²⁺ cations coordinated by 6 or 10 O²⁻ anions (Figure 1a)^{30–34}. When considering doping of the structure with manganese ions, it is expected that they will occupy V⁵⁺ sites due to the small difference in the effective ionic radii of Mn⁵⁺ and V⁵⁺ at the tetrahedral site (33.0 and 35.5 pm, respectively) and similarity in the ionic charge. On the other hand, co-doping with Nd³⁺ ions is possible due to their occupation of the crystallographic sites of Ba²⁺. Due to the lack of a reported value for the ionic radius of Nd³⁺ at the 10-fold coordinated site, it can be assumed that the Nd³⁺ ions preferentially substitute Ba²⁺ at the octahedral sites. This is also expected due to the smaller difference between the ionic radii of Nd³⁺ and Ba²⁺ located at the 6-fold coordinated site than with that located at the 10-fold coordinated site (98, 3, 142 and 152 pm, respectively). However, it should be noted that there is a charge mismatch between the Nd³⁺ and Ba²⁺ ions, which will generate additional defects, and the high concentration can lead



to significant changes in the structure of the vanadate along with a high number of optical traps.

To verify whether the concentration of Nd^{3+} ions affects the crystallographic structure of the synthesized $\text{Ba}_3(\text{VO}_4)_2$ nanocrystals, the XRPD patterns were analyzed (Figure 1b). Previous work showed that the diffraction reflection of $\text{Ba}_3(\text{VO}_4)_2$ doped with 0.1% Mn^{5+} corresponds to the reference pattern (ICSD 167695) confirming the phase purity of the synthesized phosphors. Furthermore, co-doping with Nd^{3+} ions with concentrations in the range of 0.1% to 5% also does not lead to the formation of an additional crystalline phase.

Transmission electron microscopy (TEM) studies of $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}$ co-doped with Nd^{3+} ions are also in agreement with previous studies for materials singly doped with Mn^{5+} ions²⁵. It can be concluded that the particles are well crystallized with a tendency to agglomerate (Figure 1 c-f and Figure S1). The particle size distribution determined by the Feret's method suggests that more than 90% of the observed particles are smaller than 100 nm (Figure S2). For materials co-doped with 0.1% Nd^{3+} , the average particle size is estimated to be 65 nm, while when the Nd^{3+} concentration increases to 5%, the particle size increases to 84 nm. However, some sub-microcrystalline particles can be found in the samples. This observation is in agreement with the average particle sizes of $\text{Ba}_3(\text{VO}_4)_2:0.01\% \text{Mn}^{5+}$ and $\text{Ba}_3(\text{VO}_4)_2:0.01\% \text{Mn}^{5+}$, 2% Er^{3+} previously reported with 52 nm and 72 nm in size, respectively. It also confirms the previously suggested effect of Ln^{3+} ion concentration on the increasing size of the particles. The results of the analysis of the chemical composition of the samples (ICP) are presented in Table S1 and are in agreement with the stoichiometric amounts used for the synthesis.



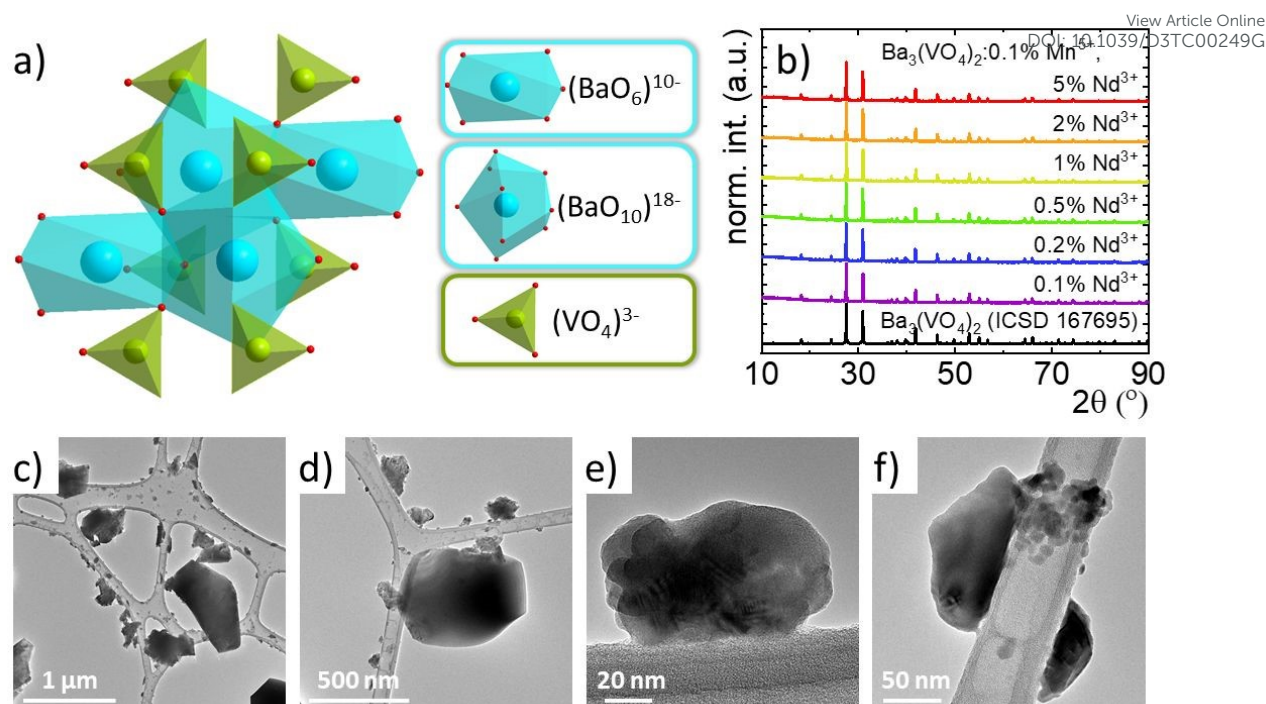


Figure 1. Structural characterization of synthesized materials: Crystal structure of the $\text{Ba}_3(\text{VO}_4)_2$ unit cell- a); X-ray diffraction patterns of $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}$ co-doped with different Nd^{3+} concentration- b); the representative TEM images of $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}$, $0.1\% \text{Nd}^{3+}$ materials- c-f).

Characterization of luminescence properties

To understand the spectroscopic properties of the $\text{Mn}^{5+}/\text{Nd}^{3+}$ co-doped system, the simplified configurational coordinate diagram of Mn^{5+} with the energy diagram of Nd^{3+} is presented in Figure 2a. In the excitation spectra of the $\text{Ba}_3(\text{VO}_4)_2:\text{Mn}^{5+},\text{Nd}^{3+}$ nanocrystals two broad absorption bands can be found with the maxima at around 370 and 700 nm. The most intense latter one corresponds to the electronic transition from the $^3\text{A}_2$ ground state to the $^3\text{T}_2$ excited state. Since $\lambda_{\text{exc}} = 808 \text{ nm}$ is a relatively inexpensive light source, matches with the $^3\text{A}_2 \rightarrow ^3\text{T}_2$ absorption band, and falls within the NIR-I spectral range, it was used as an excitation source in this work. The population of the excited state is followed by a non-radiative depopulation to the excited emitting state ^1E . The $^1\text{E} \rightarrow ^3\text{A}_2$ radiative transition leads to the generation of the narrow emission band at approximately 1180 nm. As already proved¹⁶ the



most sensitive optical response in the case of the TM/Ln³⁺ co-doped luminescence thermometers can be achieved when the employed irradiation excites independently both TM and Ln³⁺. This is because when the TM→Ln³⁺ energy transfer is responsible for the population of the excited state of Ln³⁺, it affects its luminescence thermal stability and thus lowers the sensitivity of the luminescence thermometer. In this sense, Nd³⁺ is a promising candidate since its intense absorption band corresponding to the ⁴I_{9/2}→⁴F_{5/2}, ²H_{9/2} transition overlaps with the 808 nm excitation chosen for this study. Upon this excitation, the ⁴F_{5/2}, ²H_{9/2} state is populated and its nonradiative depopulation causes the occupation of the ⁴F_{3/2} metastable state. As a result of the radiative depopulation of the metastable state, three emission bands occur at 880, 1060, and 1350 nm corresponding to the ⁴F_{3/2}→⁴I_{9/2}, ⁴F_{3/2}→⁴I_{11/2} and ⁴F_{3/2}→⁴I_{13/2} electronic transitions.

To evaluate the theoretical assumptions and verify the existence of potential energy transfers between Nd³⁺ and Mn⁵⁺ ions, the excitation spectra recorded at 83 K were analyzed (Figure 2b). In the case of the excitation spectra of Nd³⁺ ions monitored at λ_{em} = 1062 nm, numerous narrow bands have been noted which can be assigned to the transitions from the ⁴I_{9/2} level to the ⁴F_{3/2} (839 nm), ⁴F_{5/2}, ²H_{9/2} (806.5 nm), ⁴F_{7/2}, ⁴S_{3/2} (751 nm), ⁴F_{9/2}, ²H_{11/2} (680 nm), ⁴G_{5/2} (580 nm), ²G_{7/2} (575 nm), ⁴G_{7/2} (526.5 nm), ²K_{13/2}, ⁴G_{9/2}, ²K_{13/2} (510 nm), ²G_{9/2} (476.5 nm), ²K_{15/2}, ⁴G_{11/2} (448 nm), ²P_{1/2} (429.5 nm), ⁴D_{3/2}, ⁴D_{5/2} (358.5 nm) and ⁴D_{1/2} (347 nm) state. Furthermore, the spectra show two broad overlapped bands with maxima at 340 nm and ~305 nm that come from the charge transfer from the (VO₄)³⁻ group^{25,30}. The band centered at 340 nm decreases markedly compared to the band at ~305 nm as the concentration of Mn⁵⁺ increases, while the maximum of the second band gradually shifts from 312 nm to 301 nm for the phosphor doped with 0.1% and 5% of Mn⁵⁺ ions. This may be related to the gradual decrease in the strength of the crystal field that interacts with the (VO₄)³⁻ groups resulting from the substitution of larger Ba²⁺ ions with smaller Nd³⁺ ions. Due to the absence of additional bands



in the excitation spectrum of Nd^{3+} , the energy transfer from Mn^{5+} to Nd^{3+} was excluded. A different shape is presented by the excitation spectrum of Mn^{5+} measured at $\lambda_{\text{em}} = 1177 \text{ nm}$ at 83 K (Figure 2c). It consists of broad excitation bands of Mn^{5+} ions originating from electron transitions ${}^3\text{A}_2 \rightarrow {}^1\text{A}_1$ (772 nm) and ${}^3\text{A}_2 \rightarrow {}^3\text{T}_1({}^3\text{F})$ (the broadband from 590 to 750 nm with a maximum at $\sim 684 \text{ nm}$), as well as two bands associated with charge transfer of the $(\text{VO}_4)^{3-}$ group centered at 344 nm and 300 nm. As in the case of the Nd^{3+} excitation spectra, there was a marked decrease in the intensity of the band at 344 nm and its shift from 344 nm to 340 nm as the concentration of Nd^{3+} ions increased compared to the band at 300 nm. It is noteworthy that for Nd^{3+} concentration above 1% additional narrow bands occur, among others, around 805 nm, 582 nm, and 575 nm, corresponding to the previously described Nd^{3+} absorption bands. This result confirms the $\text{Nd}^{3+} \rightarrow \text{Mn}^{5+}$ energy transfer for higher dopant concentrations. The independence of simultaneous excitation of both Mn^{5+} and Nd^{3+} was confirmed by the analysis of the emission spectra of $\text{Ba}_3(\text{VO}_4)_2:\text{Mn}^{5+}, \text{Nd}^{3+}$ upon excitation at 808 nm. Even for low Nd^{3+} concentrations three emission bands at 1060, 1180 nm and 1350 nm can be observed (Figure 2d). For higher Nd^{3+} concentrations, a small spectral shift of the Mn^{5+} band maximum occurs from 1174 to 1180 nm. This shift likely stems from an increased crystal field strength affecting the Mn^{5+} sites with a growing number of Nd^{3+} ions and it is related to a smaller ionic radius of the dopant in respect to Ba^{2+} ions. A unique feature of Mn^{5+} compared to other TM ions is its extremely narrow FWHM of 2.2 nm for the sample with 0.1% Nd^{3+} ions. The FWHM value gradually increases up to $\sim 8.7 \text{ nm}$ for $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{ Mn}^{5+}, 5\% \text{ Nd}^{3+}$, which may be related to the formation of additional asymmetries in the tetrahedral site of Mn^{5+} ions due to defects introduced by the electronic charge mismatch between Ba^{2+} and Nd^{3+} ions. Besides, vibronic sidebands of Mn^{5+} ions at 1168.8 nm, 1222.2 nm and 1291.3 nm can be noted. For nanocrystals doped with 0.1% and 0.2% Nd^{3+} , weak bands at 1075 nm and 1062 nm corresponding to the Stark levels of the ${}^4\text{F}_{3/2} \rightarrow {}^4\text{I}_{11/2}$ transition are also detectable. Interestingly, above 0.5% Nd^{3+}



these bands become more intense comparing the spectra normalized to the Mn^{5+} band maximum, and additional bands at 1330 nm associated with the $^4F_{3/2} \rightarrow ^4I_{13/2}$ transition and additional Stark sublevels of the $^4F_{3/2} \rightarrow ^4I_{11/2}$ transition at 1105 nm and 1092 nm appear.

To quantify the increase in emission intensity of Nd^{3+} ions with respect to Mn^{5+} emission, the luminescence intensity ratio (LIR) parameter was introduced according to equation (Eq.1):

$$LIR = \frac{\int_{1077nm}^{1057nm} I(^4F_{3/2} \rightarrow ^4I_{11/2})[Nd^{3+}]d\lambda}{\int_{1195nm}^{1168nm} I(^1E \rightarrow ^3A_2)[Mn^{5+}]d\lambda} \quad (\text{Eq. 1})$$

The LIR value normalized to the value obtained for 0.1% Mn^{5+} , 0.1% Nd^{3+} plotted as a function of Nd^{3+} concentration (Figure 2e) showed a value of 20 at 0.5% Nd^{3+} , reaching a maximum of 43 for 1% Nd^{3+} , before slightly decreasing to 37 for 5% Nd^{3+} . This trend at higher concentration is probably associated with the reduction of the Nd^{3+} emission intensity via $\{^4F_{3/2}, ^4I_{9/2}\} \leftrightarrow \{^4I_{15/2}, ^4I_{13/2}\}$ cross-relaxation process. The concentration quenching of the intensity of the Nd^{3+} emission is confirmed by the emission spectra recorded for $Ba_3(VO_4)_2$ co-doped with 1% Mn^{5+} and various concentrations of Nd^{3+} ions (in the range of 0.1% to 10%), where Nd^{3+} luminescence could not be observed (Figure S3).



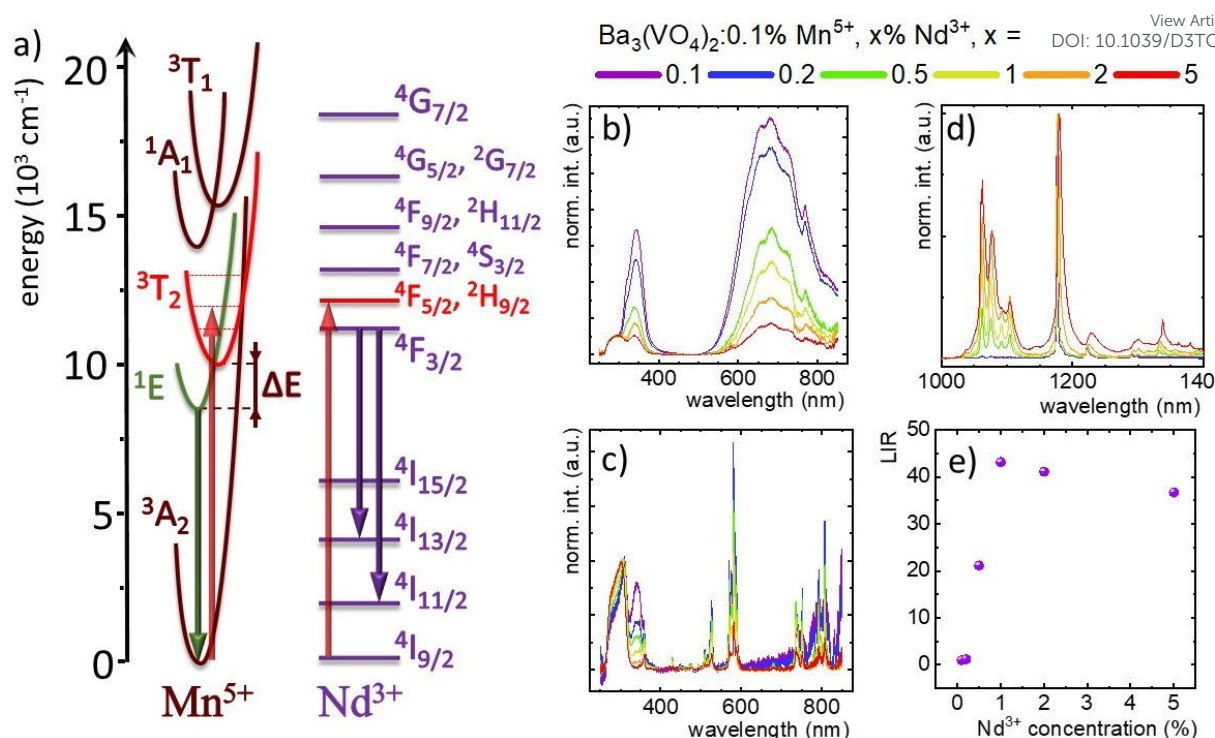


Figure 2. Configurational coordinate diagram for Mn^{5+} and Nd^{3+} ions - a); Comparison of the low-temperature (83 K) excitation monitored at Mn^{5+} emission ($\lambda_{\text{em}} = 1177 \text{ nm}$) - b) and at Nd^{3+} emission ($\lambda_{\text{em}} = 1062 \text{ nm}$) - c); Emission spectra ($\lambda_{\text{exc}} = 808 \text{ nm}$) of $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}$ co-doped with different Nd^{3+} concentration- d); Influence of Mn^{5+} concentration on the LIR parameter- e).

Thermometric properties of $\text{Mn}^{5+}/\text{Nd}^{3+}$ co-doped system

The thermal dependence of a series of materials doped with 0.1% Mn^{5+} and different concentrations of Nd^{3+} ions was analyzed (Figure 3a and Figure S4, S5, S6). In the case of low Nd^{3+} concentration, the Mn^{5+} intensity initially increases up to around 210 K followed by the decrease in the intensity. Hence the T_{50} (temperature at which the emission intensity reached half of its initial intensity at low temperature) is higher for the Mn^{5+} , 0.1-1% Nd^{3+} co-doped samples ($T_{50} \sim 380\text{-}390 \text{ K}$) compared to the solely Mn^{5+} doped counterpart. Further increases in the Nd^{3+} concentration hamper the initial enhancement of emission intensity and thus the T_{50} decreases to 270 K for 2%. This initial increase in emission intensity was already reported for



materials doped with Cr³⁺ and Mn⁴⁺ ions and was explained in terms of the overlap of the wavefunctions of two states and thus thermal increases in the probability of radiative depopulation of the emitting state^{35–39}. On the other hand, the Nd³⁺ emission intensity was expected to be affected to a lower extent by temperature changes (Figure 3c). For 0.1% Nd³⁺, a slight increase in intensity was observed up to 423 K. Due to the low emission intensity of the Nd³⁺ ion for low dopant concentration, the integral emission intensity may be affected by the strong contribution of the Mn⁵⁺ emission intensity. For a higher amount of dopant (0.5% and 1% Nd³⁺) the monotonic decreases of emission intensity at elevated temperature up to up to 25% at 683 K was found. Nevertheless, the Nd³⁺ was found to be significantly less susceptible to thermal changes with respect to the Mn⁵⁺, which enables the development of a luminescence thermometer based on the ratio of their emission intensities (Figure 3d). The LIR value for 0.1% Mn⁵⁺, 0.1% Nd³⁺ co-doped nanocrystals in the 83 - 243 K temperature range remains almost unchanged. Above 243 K the quasi-exponential increase in the LIR value up to 22.0 at 703 K can be observed. Interestingly, when the concentration of Nd³⁺ increases, some saturation or even a decrease in the thermal dependence of LIR can be found. The temperature at which the reversion of the LIR monotonicity can be observed decreases with the dopant concentration from 690 K for 0.5% Nd³⁺ to 580 K for 5% Nd³⁺. This is expected since at high temperatures the emission intensity of the Mn⁵⁺ is almost completely quenched and the thermal change of Nd³⁺ emission intensity reversed the thermal variation of the LIR. To verify the influence of Mn⁵⁺ concentration on the thermal changes of the LIR values quantitatively, the relative sensitivity S_R was calculated according to the following formula (Eq. 2):

$$S_R = \frac{1}{LIR} \frac{\Delta LIR}{\Delta T} \times 100\% \quad (\text{Eq. 2})$$

where ΔLIR represents the change of LIR for the ΔT change of temperature. The obtained S_R results were plotted in Figure 3e. For 0.1% Nd³⁺, the maximum sensitivity S_{Rmax} of about 0.83% K⁻¹ was obtained at ~433 K, with $S_R = \sim 0.40\%$ K⁻¹ at 303 K. For 0.2% Nd³⁺, the value of S_{Rmax}



= 0.88% K⁻¹ was obtained at the highest temperature among the analyzed materials, i.e. 483 K. For higher concentrations of Nd³⁺ ions, a gradual shift in T(S_{Rmax}) toward lower temperatures was observed. The highest sensitivity value was obtained for 0.5% Nd³⁺, which was equal to S_{Rmax} = 0.94% K⁻¹ at 453 K. The highest S_R ~0.69% K⁻¹ in the physiological temperature range was featured by the Ba₃(VO₄)₂:0.1% Mn⁵⁺, 2% Nd³⁺ sample.

To verify the applicative potential of Ba₃(VO₄)₂:0.1% Mn⁵⁺, 2% Nd³⁺ as a luminescent thermometer, an experiment with light-induced heating while simultaneously obtaining a thermal readout was performed. The excitation power density of the λ_{exc}=808 nm was changed and the luminescence spectra of Mn⁵⁺ and Nd³⁺ were measured. In addition, the temperature of the sample was controlled by using an infrared thermometric camera. As can be seen, the increase in excitation density results in the increase in the intensity of the Nd³⁺ emission, as expected (Figure S7). On the other hand, the emission intensity of Mn⁵⁺ ions decrease at a lower rate compared to the previously described experiments (Figure 3a). This stems from the fact that luminescence thermal quenching competes with the increase of the emission intensity associated with the increase in the excitation density. The comparison of the temperature determined by the thermal camera with the temperature determined by luminescence thermometry (LIR) in Figure 3f reveals a very good agreement in the 300-450 K temperature range. This result confirms the high application potential of Ba₃(VO₄)₂:0.1% Mn⁵⁺, 2% Nd³⁺ for temperature sensing.

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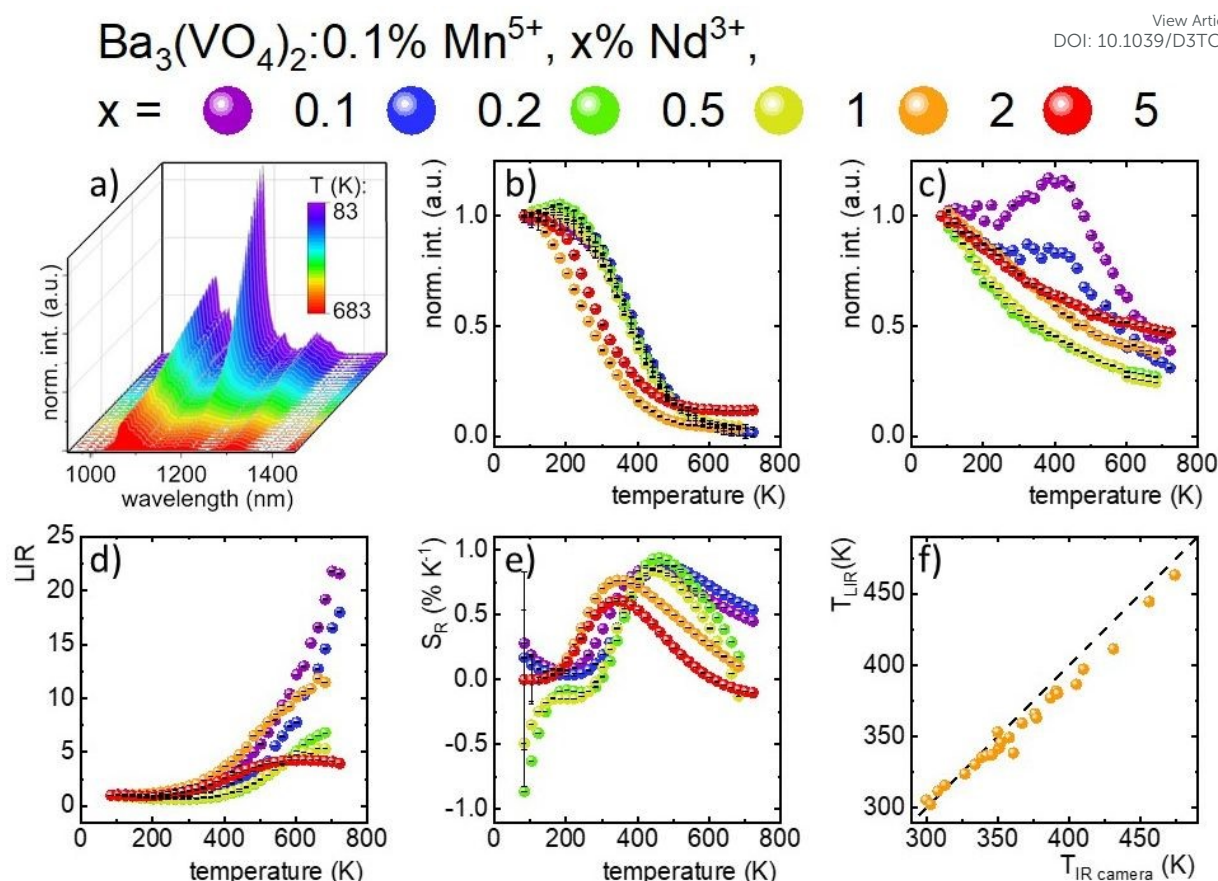


Figure 3. Thermal evolution of the emission spectra of $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}, 2\% \text{Nd}^{3+}$ phosphor- a); Thermal evolution of Mn^{5+} band intensity- b); Nd^{3+} band intensity- c); LIR- d) and S_R of $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}$ co-doped with different concentrations of Nd^{3+} - e); Comparison of temperature calculated with LIR and temperature detected by IR camera using $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}, 2\% \text{Nd}^{3+}$ - f).

Thermal imaging based on Mn^{5+} and Nd^{3+} ratio

Promising temperature sensing results based on LIR of Mn^{5+} and Nd^{3+} prompted testing of the developed material's thermal imaging prowess and corresponding thermometric approach. Figure S10 shows a schematic representation of the measurement system, which includes a laser diode with an 808 nm excitation line, an NIR spectral camera, and a thermal camera. The phosphor powder was introduced between two glass microscope slides and one edge of the slide was placed in contact with a cold source. By heating the other edge of the glass slide, a temperature gradient was created. In order to provide ratiometric readout the 2D images



of luminescence were measured using alternately two optical filters i.e. a 1200 nm bandpass filter with FWHM = 25 nm (for Mn^{5+} luminescence) and 1150 nm shortpass filter (for Nd^{3+}) integrated in neighboring positions in an automated filter wheel. Representative intensity maps recorded during this experiment are shown in Figures 4a (Mn^{5+}) and 4b (Nd^{3+}). Near the cooling area the intensity values of both images are similar. However, as the heating source was applied, the intensity in both maps decreased, but at a faster rate for the map in Figure 4a. This difference is manifested in the LIR map, obtained by dividing the image taken with the 1200 nm bandpass filter by the image taken with the 1150 nm shortpass filter (Figure 4c). The change in LIR values as a function of distance from the heating source can be clearly observed. It should be noted that lower LIR values were obtained near the heating source, indicating an inverse relation with the temperature gradient. Yet, to obtain a temperature map from the LIR values, it was necessary to perform a calibration curve (Figure 4d). Therefore, a series of measurements at predetermined constant temperatures of a heating plate were carried out. The resultant intensity images obtained for the two ions (Mn^{5+} and Nd^{3+}) were divided by each other to obtain the LIR images. Finally, a linear equation was used to convert the LIR map into a temperature map, shown in Figure 4e. The resulting map shows a temperature gradient that covers a range of about 280 - 400 K, which agrees with the temperature gradient obtained from the thermal camera (Figure S11).

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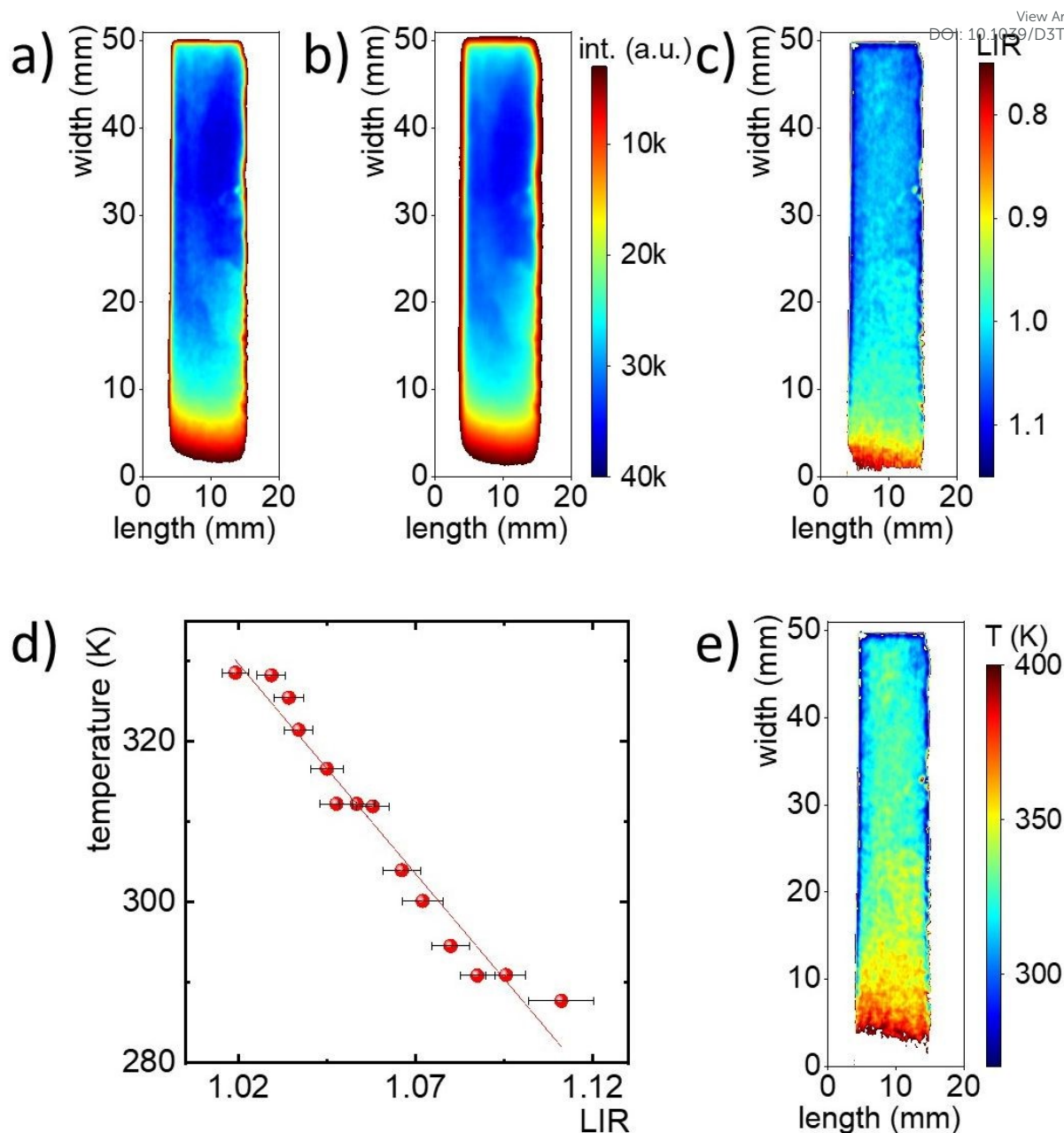


Figure 4. The proof of concept experiment with an induced temperature gradient for $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}$, 2% Nd^{3+} phosphor: the intensity map with a 1200 nm bandpass filter- a), with a 1150 nm shortpass filter- b); the LIR map- c); the calibration curve of temperature vs LIR- d) and the calculated temperature map- f).

Ratiometric thermal imaging through tissue phantom

Since the spectral range of the Mn^{5+} and Nd^{3+} emission fall within the NIR-II spectral range, one would consider $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}$, 2% Nd^{3+} suitable for through-tissue imaging and thermal sensing. To test this capability, the thermal mapping performance was further



evaluated through tissue phantoms with different compositions (A and B, respectively without and with ink, alongside Intralipids). The phantom composition was selected to study the effect of intralipid concentration, which mainly enhanced the scattering properties and Indian ink, which was responsible for improving the absorption properties on the total extinction properties. In each case, the phantom was placed on a glass slide, under which the Mn^{5+} , Nd^{3+} co-doped phosphor was placed (Figure S10).

The same measurements as described above were performed using a 1200 nm bandpass filter and a 1150 nm shortpass filter (Figure S11). In the case of the ink-free phantom, the intensity scale in which the two images are located differs significantly. Moreover, in the high temperature area, the intensity of the Mn^{5+} image dropped dramatically, while for the Nd^{3+} intensity image the values were similar throughout the measurement area. By dividing the fluorescence images by each other, an LIR map was obtained, which clearly shows the temperature gradient (Figure 5b). However, the range of LIR values for which the temperature gradient is observed remains around 0.3 - 0.7, which significantly deviates from the LIR range in which the temperature gradient was recorded in the absence of tissue phantom (i.e., LIR between ~0.8 and 1.1). This indicates that the previously performed calibration curve recorded in the absence of the phantom is not appropriate and cannot be applied in this case. In addition, a similar experiment was performed using tissue phantom B (Figure S12), which also contained Indian ink. As shown in the filter images, in both cases the temperature gradient is not distinguishable (Figure S13). This is also confirmed by the LIR maps, where the LIR values remain close to each other in the central part of the sample at around 0.6 (Figure 5c).

To quantitatively compare the differences obtained from the three LIR maps, individual profiles marked as white dashed lines in Figure 5a-c, which are consistent with the direction in which the temperature gradient was generated, were selected and contrasted as a function of the



distance from the heating element (Figure 5d). The LIR profiles calculated for the cross sections of samples with and without phantoms differ significantly.

Therefore, the luminescent thermometer was re-calibrated in the presence of tissue phantom A, acquiring luminescence images as a function of temperature while monitoring the temperature by the thermal camera. The so-acquired calibration curve was compared with the one determined for phosphor without optical obstructions (Figure 5e). Unquestionably, the two calibration curves differ in the range of LIR values that cover the temperature range of 280 to 330 K. In addition, it can be found that due to the lower signal to noise ratio observed in this case the noisy character of the temperature profile when measured through tissue phantom A. Despite this, a linear function was determined that could serve as a calibration curve for the LIR vs temperature values. It resulted in the determination of a temperature map that confirms the temperature gradient obtained through tissue phantom A (Figure S14). Such a dramatic difference in the temperature readout of the same system with and without a tissue phantom forecloses the use of the ratiometric approach in bio-related thermal imaging unless specific actions are undertaken to obviate this limitation⁴⁰. The effect described above is a consequence of the different extinction at different wavelengths and for different tissues. As it turns out in the example of tissue phantom A, the difference in LIR stems from the different value of μ_{ext} for the spectral ranges that covered the filters used in the experiment, i.e. $\mu_{\text{ext}} = 13.45 \text{ cm}^{-1}$ and $\mu_{\text{ext}} = 13.65 \text{ cm}^{-1}$ for wavelengths of 1070 nm and 1180 nm, respectively (Figure 5f). On the other hand, the tissue phantom B failed to register a temperature gradient, due to an excessively high extinction coefficient exceeding 16.0 cm^{-1} over the entire spectral range. This is a consequence of the high scattering coefficient, the increase of which plays a dominant role after the addition of Indian ink with carbon particles in relation to the absorption coefficient.

In conclusion, the developed $\text{Mn}^{5+}/\text{Nd}^{3+}$ nanophosphor presents good performance for temperature sensing. However, in experiments conducted with tissue phantoms, it is evident



that it cannot be immediately translated, as is in deep-tissue thermal imaging based on the LIR approach. This is due to the fact that the Mn^{5+} emission band partially overlap with the absorption overtone of water molecule. Conversely, the lifetime-based approach discussed in our previous work enables to provide a reliable temperature readout even in deep tissue imaging. Therefore, the presented results confirm the superiority of the luminescence kinetics-based approach over the ratiometric approach for through-tissue thermal imaging.

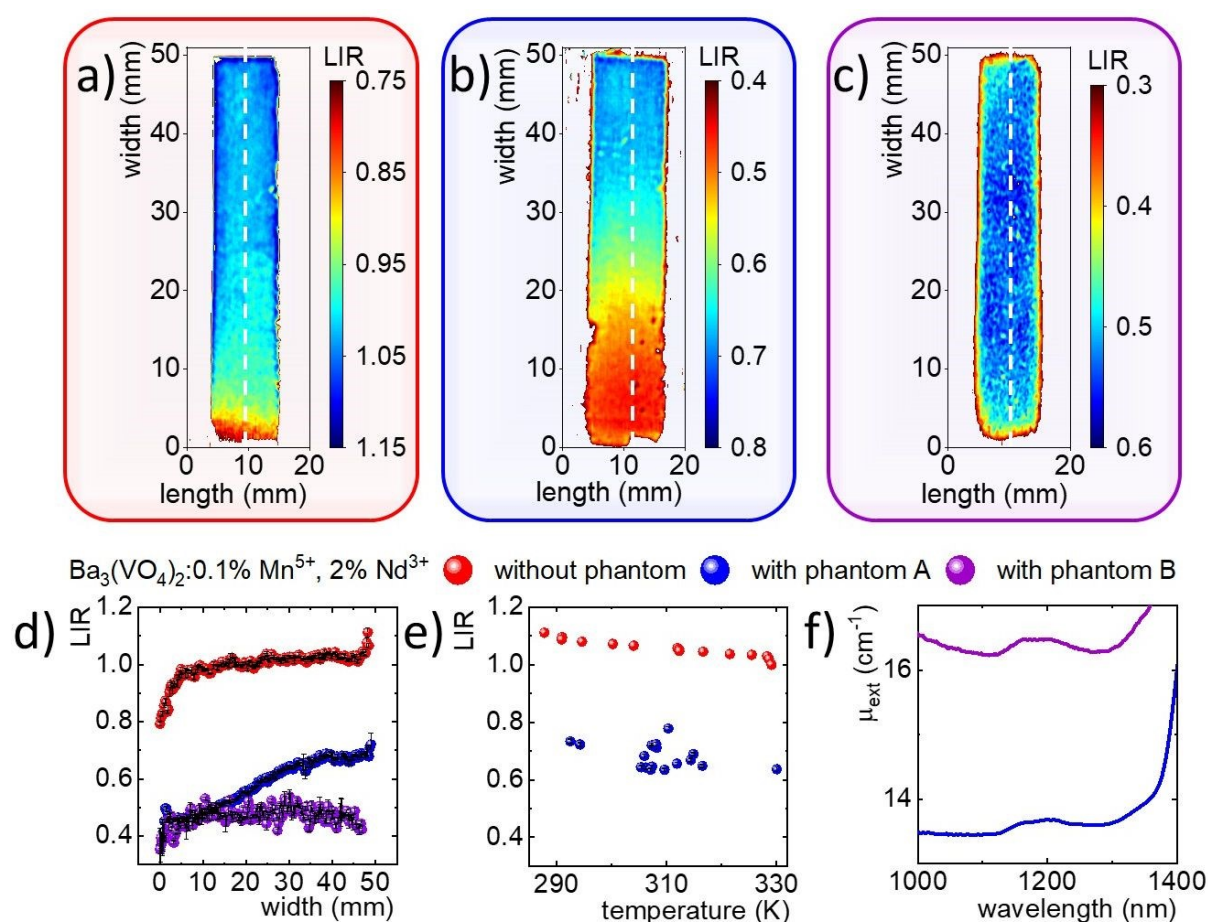


Figure 5. LIR maps with induced temperature gradient for $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}, 2\% \text{Nd}^{3+}$ phosphor without tissue phantom – a), with tissue phantom A – b) and with tissue phantom B – c); comparison of LIR values in the function of distance from the heating source – d); comparison of calibration curves of LIR vs temperature for $\text{Ba}_3(\text{VO}_4)_2:0.1\% \text{Mn}^{5+}, 2\% \text{Nd}^{3+}$ phosphor determined directly or through tissue phantom – e); extinction coefficient as a function of wavelength for tissue phantom A and B – f).

Conclusions



In the present study, the thermometric properties of a ratiometric luminescent thermometer based on the emission intensities ratio of ${}^1E \rightarrow {}^3A_2$ of Mn^{5+} and ${}^4F_{3/2} \rightarrow {}^4I_{11/2}$ Nd^{3+} were analyzed as a function of Mn^{5+} and Nd^{3+} ions concentration. The emission intensity ratio of Mn^{5+} to Nd^{3+} ions due to inter-ionic energy transfer can be modulated by the concentration of Mn^{5+} ions. The difference in the electron configuration of the considered dopant ions generates a difference in the temperature quenching rate of their luminescence and thus makes it possible to use their LIR as a thermometric parameter. As the concentration of Nd^{3+} ions increased, the temperature range for which maximum sensitivities were recorded gradually decreased. Therefore, $Ba_3(VO_4)_2:0.1\% Mn^{5+}, 2\% Nd^{3+}$ was chosen as the optimal temperature sensor in the physiological range. Confirmation of the application potential of the developed thermometer was provided by remote temperature measurements during laser-induced heating of the powder sample. However, the significant difference in extinction coefficients occurring at different measurement wavelengths in the case of thermal imaging using a tissue phantom showed that the described problems must be eliminated to increase the reliability of the measurement in such experiments. The presented work was focused on the fundamental studies of luminescence-based temperature imaging using $Ba_3(VO_4)_2:0.1\% Mn^{5+}, 2\% Nd^{3+}$ and issues with relatively wide particle size distribution and particle aggregation must be resolved before real application. In addition, the influence of the bio-related medium on the luminescence response of this temperature sensor should be verified.

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